

Problem Solving

Buck

$$V_s = 50 \text{ V}$$

$$D = 0.4$$

$$L = 400 \text{ } \mu\text{H}$$

$$C = 100 \text{ } \mu\text{F}$$

$$f = 20 \text{ kHz}$$

$$R = 20 \text{ } \Omega$$

Assuming ideal components, calculate (a) the output voltage , (b) the maximum and minimum inductor current, and (c) the output voltage ripple.

Boost

A boost converter has following parameters. Determine duty cycle, average, minimum and maximum inductor current and output voltage ripple.

$$V_{IN} = 12 \text{ V}$$

$$V_{OUT} = 30 \text{ V}$$

$$L = 120 \text{ } \mu\text{H}$$

$$C = 48 \text{ } \mu\text{F}$$

$$R = 50 \text{ ohm}$$

$$f = 50 \text{ KHz}$$

Buck-Boost

$$V_s = 24 \text{ V}$$

$$D = 0.4$$

$$R = 5 \, \Omega$$

$$L = 20 \, \mu\text{H}$$

$$C = 80 \, \mu\text{F}$$

$$f = 100 \text{ kHz}$$

Determine the output voltage, inductor current average, maximum and minimum values, and the output voltage ripple